OAKLAND UNIVERSITY TEMPLATE FOR LATEX BASED MASTERS THESIS AND PHD DISSERTATIONS

DISSERTATION FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY IN ELECTRICAL AND COMPUTER ENGINEERING

STEVEN LOUIS

OAKLAND UNIVERSITY

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OAKLAND UNIVERSITY TEMPLATE FOR LATEX BASED MASTERS THESIS AND PHD DISSERTATIONS

by

STEVEN LOUIS

A dissertation submitted in partial fulfillment of the
requirements for the degree of

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Oakland University
Rochester, Michigan

Doctoral Advisory Committee:

Hari Seldon, Ph.D., Chair
Philip Brainard, Ph.D.
Henry Jones, Jr., Ph.D.
James Moriarty, Ph.D.

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To my beloved,

Without you, this project would have gotten nowhere.

Please change this text in the "0000dedication.tex" file to

be your own dedication.

ACKNOWLEDGMENTS

The author wishes to express his appreciation for extensive mentorship provided by Hari Seldon, my adviser. Thanks are also due to committee members Philip Brainard, Ph.D., Henry Jones, Jr., Ph.D., James Moriarty, Ph.D. for their contributions to this dissertation. Likewise, acknowledgment should be extended to ...

Please change this text in the “000acknowledgements.tex” file.

Steven Louis

ABSTRACT

OAKLAND UNIVERSITY TEMPLATE FOR LATEX BASED MASTERS THESIS AND PHD DISSERTATIONS

by

Steven Louis

Adviser: Hari Seldon, Ph.D.

This pdf might be named “Instructions.pdf”. This was generated when this latex template was put together. After the file is compiled on your computer, this pdf might be named “OUThesis.pdf”. These two pdf files should be compared, and be the same.

To write your own abstract, open the file “001abstract”, delete this text, and start typing.

This PDF was generated by a LaTeX to conform to Oakland University dissertation or thesis formatting requirements. It is hoped that by providing a set of tex files, that it will be easier to produce a long, properly formatted document. The benefits of using LaTeX are many-fold; for example, the bibliography is far easier to produce, equations and figures will be automatically numbered, among other things. However, LaTeX does have a learning curve for those who have never used LaTeX before.

This document is a work in progress- if you notice any problems or have suggestions for improvement, please contact Prof. Steven Louis in the Electrical and Computer Engineering Department (SLouis@Oakland.edu).

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LIST OF ABBREVIATIONS

FPGA	Field-Programmable Gate Array

CHAPTER ONE

BASIC EDITING

1.1 Using this template

To create your chapter 1, find the file named “01Chapter.tex” and open it in a latex editor. Then, replace all of this text. This file cannot be compiled directly; compiling this text and figures into a pdf is achieved by compiling the ‘OUThesis.tex’ file. Please try to compile a duplicate of this pdf before starting. Pay particular attention to making the “List of Abbreviations” in section 1.6

1.2 Text Editing

In Microsoft Word, typing occurs in paragraphs. For proof reading, locating the mistakes in their paragraph and correcting them in Microsoft Word is relatively easy. In LaTeX, however, correcting the mistakes during proof reading can at times a little difficult. To make it easier, it is suggested to write each sentence on its own line. This way, when there is an error to correct, one can simply look at the first word of the sentence to locate the text that should be changed.

1.3 The OUThesis.tex File

As stated earlier, to compile your dissertation, the ‘OUThesis.tex’ file should be compiled. When starting, there are a number of things that should be adjusted in that file when work is begun. There are comments in the OUThesis.tex file to explain this.

The preamble is the portion of the file from top to ‘\begindocument’. The following should be adjusted in the preamble:

- documentclass: If you are a PhD student writing your dissertation, the document class does not need to be changed. If you are a masters student, the document class should be changed to read '\documentclass[thesis, bookmarks]{grizz_thesis}'.

- includeonly: The '\includeonly' command can be uncommented to compile a subset of your document.

- usepackage: If you wish to use additional packages, they can be added to the preamble. The packages already in use can be found in the 'grizz_thesis.cls' file.

- custom macros: For a long thesis, it may be a good idea to use macros to make typing it in easier. For example, ω and $v_{\text{stno}}(t)$ have macros defined.

- newacronym: The '\newacronym' part of the preamble will be discussed in section 1.6.

- year: The year in your document can be changed in the preamble.

Between '\begin{document}' and '\end{document}' in the 'OUThesis.tex' file, there are some things that need to be changed. The main thing is to enter the title, name, degree, and committee names. Besides this, the additional chapters and appendices should be uncommented as your thesis grows.

1.4 Equations

The beauty of latex is that if is excellent for formatting equations, and once you have the syntax memorized, equations are fast to type. For example, the equation $\alpha(x) = \beta\delta y\sqrt{\varepsilon}$ is typed inline between two $ signs. If instead an equation should be numbered, it might be best to code it as:

$$\alpha(x) = \beta\delta y\sqrt{\varepsilon} \tag{1.1}$$

To have an equation without numbering, simply use the following:

$$\alpha(x) = \beta\,\delta y\sqrt{\varepsilon}$$

How, one might ask, should an equation be referenced in the text? The easy way is to use the included package, "hyperref". This allows you to refer to Equation 1.1 and Equation 1.3a and Equation 1.3b. As will be shown in the next section, "\autoref{ }" also works with figures and tables and the like. Please note that despite their black color, hyperref generates hyperlinks in the pdf output. Thus, if you click on this: Equation 1.1 your pdf viewer might take you back a page.

If you want a piecewise equation with a single number, here is an example.

$$r(t) = \begin{cases} A\cos(\pi\rho t^2 + \omega t + \psi_r) & -\frac{T}{2} \leq t \leq \frac{T}{2} \\ 0 & \text{otherwise} \end{cases} \tag{1.2}$$

If instead the pieces should be referenced individually, try:

$$v(t) = \begin{cases} -\frac{1}{2}I\Delta R\cos(\psi) & \text{when injection locked} \quad (1.3a) \\ -\frac{1}{2}I\Delta R\cos(\pi\rho t^2 + \omega t + \psi) & \text{otherwise} \quad (1.3b) \end{cases}$$

1.5 Quotations

When you have a long quotation, it may require a quotation block. Please refer to the OU thesis formatting guidelines to determine when a block is required.

> It was actually Bart Cameron's error and you'll have to understand about Bart Cameron. He's the sheriff at Twin Gulch, Idaho, and I'm his deputy. Bart Cameron is an impatient man and he gets most impatient when he has to work up his income tax. You see, besides being sheriff, he also owns and runs the general store, he's got some shares in a sheep ranch, he does a bit of assay work, he's got a kind of pension for being a disabled veteran (bad knee) and a few other things like that. Naturally, it makes his tax figures complicated.

This quote was taken from an Issav Assimov story, *The Dead Past* [1].

1.6 Acronyms

In the front matter, there is a page called “List of Abbreviations”. How was this generated?

First, it is important to discuss the ‘glossaries’ package. The glossaries package is included in the grizz_thesis.cls file. This allows the use of the \gls{} and \glspl{} commands. In the preamble of the ‘OUThesis.tex’ file, there is an line \newacronym{fpga}{FPGA}{Field-Programmable Gate Array}. This can be referenced in singular with the code \gls{fpga }, and with this command it will display as Field-Programmable Gate Array (FPGA). As can be seen, the first time \gls{fpga } is used it displays the full name and the acronym in parenthesis. The second time this command is used, it will just say FPGA. If is should be plural, it must use the code \glspl{}. Then, it will be displayed as FPGAs.

Once the glossaries package is being used correctly, the List of Abbreviations can be generated. This is a little complicated - it requires that perl be installed on your computer. If you are using Windows, add the directory of perl to the path. After this, run the command ‘makeglossaries OUThesis’ inside the directory of your latex files. This will generate the files required to populate the list of abbreviations. If you are using windows, there is a batch file ‘makeGloss.bat’ that can be used with a mouse in Windows. The command ‘makeglossaries.exe’ should also be in the path. As I used MiKTeX, it was found in the MiKTeX directory. There may be some trouble shooting associated with this.

Once all these steps are done, the List of Abbreviations should appear correctly in the generated pdf file.

Figure 1.1: First Chapter Figure. PDF vector image is six inches wide, and copied from Wikipedia. Figures are discussed in Chapter 2.

CHAPTER TWO

BACKGROUND

2.1 Using this template

To create your chapter 2, find the file named “02Chapter.tex” and open it in a latex editor. Then, replace all of this text. This file cannot be compiled directly; compiling this text and figures into a pdf is achevied by compiling the ‘OUThesis.tex’ file.

2.2 Figures

Figures are an important part of any dissertation. Figure 2.1 is an example figure. The caption in the code contains two parts, coded as “\caption[]{}”. Text placed between the square brackets [] goes to the ‘list of figures’ in the front matter. Text placed between the curly brackets {} appears on the page with the figure.

To get the figure to work properly, it should be the right size. Firstly, the width of the figure should be 6 inches or less. This will keep it out of the margins. The height of the figure + the height of the caption altogether must be small enough to fit within the margins. This should be monitored manually.

It is ideal to generate figures as vector images in pdf format. This will ensure infinite resolution with a small file size. If you are unsure what a vector image is, please research this on the internet. Also, when generating figures, make sure that all text in the figures uses a 12 point times font.

OU formatting guidelines dictate that figures should be placed within a page of its first mention. How can this be achieved with LaTex? Firstly, try putting the code for the figure just after it is mentioned. This is done with Figure 2.2. Generally, LaTex is able to put the figure in a good location automatically. Compile the file and see if its location is good.

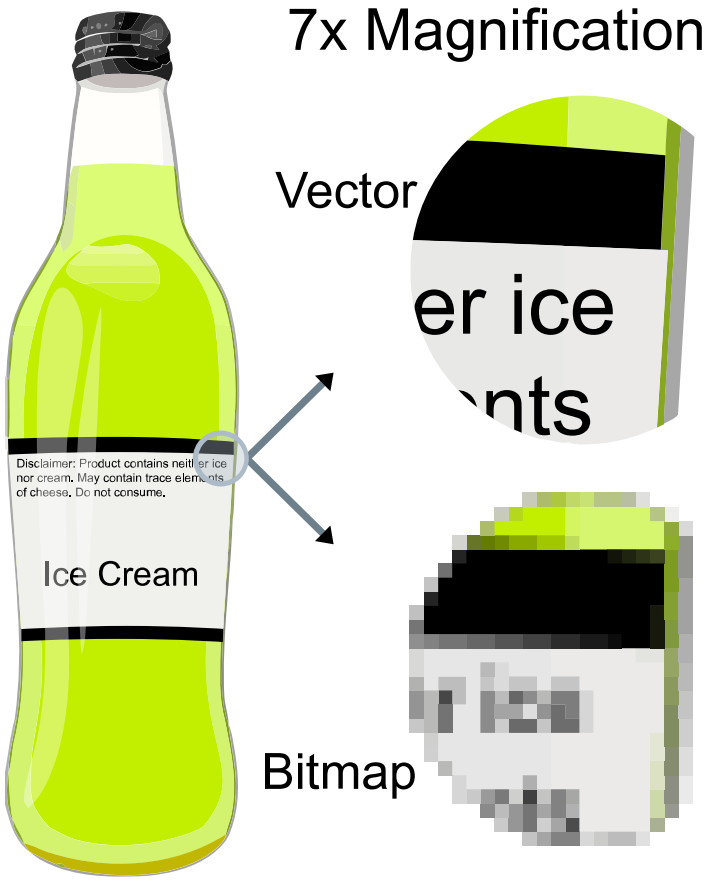

Figure 2.1: Title of figure that appears in the text. This figure is from Wikipedia, and shows the difference between a vector image and a bitmap. Note that if you zoom in all the way, the resolution is still perfect, even though the file size is just 46 kB

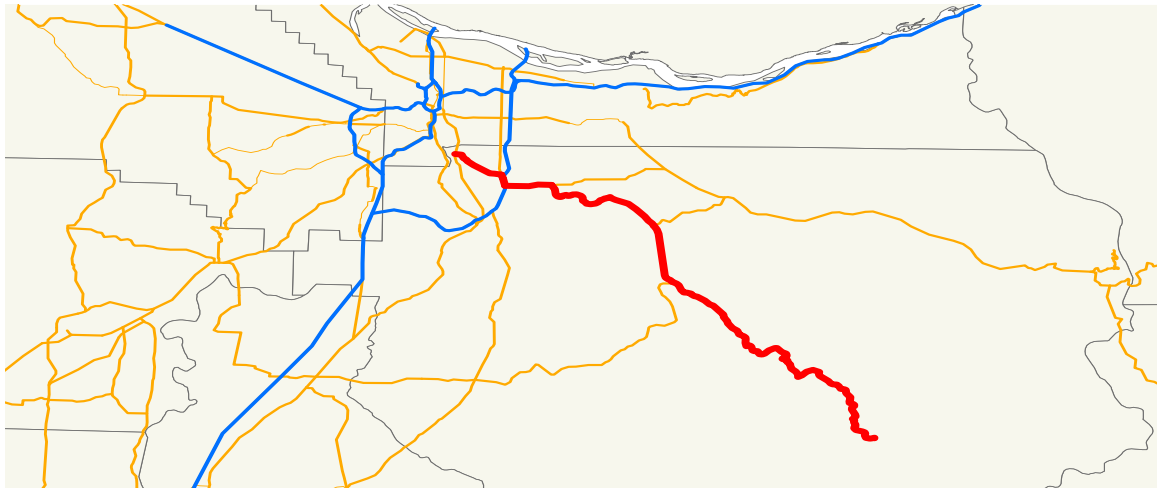

Figure 2.2: Oregon Route 224, vector image is 6 inches wide, and copied from wikipedia.

Very rarely, the location is not good. In this case, the location can be forced. As the document will change continually, only make these changes immediately before the first formatting review. To force the location, the text in the code [tbp] should be changed. If it is changed to [b] that forces the figure to the bottom of the page. If it is changed to [p] that forces the figure to have its own page.

2.3 Code Figures

There are times when code, specifically programming code, will be included in this thesis. Perhaps in the methods chapter? Perhaps elsewhere? As an example, see Figure 2.3. This may require some changes or trouble shooting.

```
# This prints Hello, world!
print('Hello, world!')
print('Hello, friends!')
```

Figure 2.3: Coding figure; Hello world in python

Table 2.1: Performance metrics for spectrum analysis

Performance Metric	**Symbol**	**Base Units**	**Brief Description**
Frequency accuracy		Hz	Accuracy with which the input signal frequency is identified.
Scanning bandwidth	δf_c	Hz	Total bandwidth to be analyzed.
Scan rate	ρ	Hz/s	The rate at which spectrum analysis is performed.

2.4 Tables

In latex, tables are efficient but also sort of a pain. Table 2.1 shows an example of a table. There a many ways to format tables; please find internet resources. The only issue is that care must be taken to make sure your table is less than 6 inches wide. One tool that can help with this is "\parbox[]{}".

CHAPTER THREE

CITATIONS

To create your chapter 3, find the file named “03Chapter.tex” and open it in a latex editor. Then, replace all of this text. This file cannot be compiled directly; compiling this text and figures into a pdf is achevied by compiling the ‘OUThesis.tex’ file.

3.1 Bibtex

This LaTeX template uses bibtex to generate the bibliography. The Bibtex file for this thesis is named ‘references.bib’. This file must be populated by sources relevant to your dissertation. How to do this? If you open the file, you can see that the entries are formatted as shown in Figure 3.1. This formatting can be downloaded directly from Google Scholar, see this website.

3.2 Citations

To cite something, simply use the ‘\cite{}’ command[2]. One can cite other things easily[3, 4]

```
@book{jeffrey2007table,
  title ={Table of integrals, series, and products, 7th ed.},
 author={Jeffrey, Alan and Zwillinger, Daniel},
 year={2007},
  publisher ={Elsevier},
 note = {Equation 2.325.13}
})
```

Figure 3.1: Coding figure; Hello world in python

APPENDIX A

EXAMPLE APPENDIX

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[1] I. Asimov, "The dead past," *Astounding Science Fiction*, vol. 81, no. 2, pp. 6–46, 1956.

[2] M. N. Sadiku and C. K. Alexander, *Fundamentals of electric circuits*. McGraw-Hill Higher Education, 2007.

[3] J. Pain and P. Rankin, *Introduction to vibrations and waves*. John Wiley & Sons, 2015.

[4] A. Jeffrey and D. Zwillinger, *Table of integrals, series, and products, 7th ed.* Elsevier, 2007. Equation 2.325.13.